

Siva Sankar

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Senior MLOps | Cloud Engineer | GenAI | Python Engineer

PROFESSIONAL SUMMARY

Results-driven MLOps and Cloud Engineer with 9+ years of progressive experience designing, automating, and deploying scalable AI, ML, and cloud-native systems across AWS, Azure, and GCP. Expert in Python backend development, DevOps automation, and Generative AI pipelines using modern frameworks such as LangChain, LangGraph, and Hugging Face.

Proven ability to bridge development and operations by implementing CI/CD, Infrastructure as Code (IaC), and container orchestration solutions using Docker, Kubernetes, Terraform, and Jenkins. Skilled at fine-tuning and deploying large language models (LLMs) and RAG architectures for enterprise-grade applications. Adept at collaborating with cross-functional teams to deliver secure, high-availability, and production-grade systems that drive business value.

Core Strengths:

MLOps • GenAI & LLMs • Cloud Infrastructure • CI/CD Automation • Python Backend • IaC (Terraform/Ansible) • Model Deployment • Vector Databases • Observability • RAG Pipelines • Cost Optimization • Performance Engineering

TECHNICAL SKILLS

Category	Technologies / Tools
Cloud Platforms	AWS (EC2, S3, Lambda, SageMaker), Azure (OpenAI, CosmosDB, Databricks), GCP
MLOps & CI/CD	SageMaker Pipelines, Azure ML, Databricks, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Ansible, Helm
GenAI & NLP	LangChain, LangGraph, GPT-4/4o, Claude, LLaMA, Hugging Face, Prompt Engineering
RAG & Vector DBs	FAISS, ChromaDB, Redis, PGVector, Pinecone, Neo4j
Programming	Python, SQL, Shell Scripting
ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, XGBoost
Backend Frameworks	FastAPI, Django, Flask
Monitoring & Observability	Prometheus, Grafana, ELK, CloudWatch
Databases & BI	PostgreSQL, MySQL, MongoDB, DynamoDB, Redshift, Power BI, QuickSight

EXPERIENCE

Compass USA | Senior MLOps & Cloud Engineer | Remote | Feb 2024 – Present

Application: AI-powered knowledge management and customer engagement platform for corporate food services.

Tech Stack: LangGraph, Claude, Python, Redis, ChromaDB, AWS, GCP, Docker, Kubernetes, Terraform, Grafana

- Designed and implemented a multi-cloud MLOps platform (AWS + GCP), reducing end-to-end deployment time by 80%.
- Built and operationalized LLM-powered GenAI workflows for document understanding, summarization, and contextual Q&A using LangChain + LangGraph.
- Developed automated RAG pipelines integrating ChromaDB and Redis for vector storage and retrieval.
- Implemented Terraform-based IaC modules for provisioning scalable model training and inference environments.
- Designed Kubernetes Helm charts for AI microservices, ensuring zero-downtime rollouts and horizontal scalability.
- Set up Grafana + Prometheus dashboards for real-time monitoring of resource utilization and model performance.
- Integrated CI/CD pipelines with GitHub Actions, automating model deployment, testing, and validation.
- Collaborated with cross-functional data science teams to productionize ML models, ensuring compliance and security.
- Optimized inference latency by 35% through container tuning and efficient caching mechanisms.

Sentara Health | DevOps & Cloud Engineer | Remote | Dec 2022 – Jan 2024

Application: Healthcare claims and clinical support platform for document automation and predictive analytics.

Tech Stack: Azure OpenAI, LangChain, PGVector, Neo4j, Docker, Kubernetes, Terraform, Jenkins

- Architected and maintained end-to-end DevOps pipelines (Jenkins + GitHub Actions) for ML and backend services.
- Deployed Azure OpenAI-based clinical assistants that processed and summarized medical claim documents.

- Developed containerized microservices for claim validation and NLP-based anomaly detection using Docker & K8s.
- Implemented Terraform for Azure IaC, achieving fully reproducible and policy-compliant infrastructure builds.
- Integrated PGVector and Neo4j for relational and graph-based data enrichment in healthcare analytics pipelines.
- Automated backup and disaster recovery processes for critical ML workloads.
- Created comprehensive monitoring and alerting solutions using Prometheus, Azure Monitor, and Grafana.
- Partnered with data scientists to containerize machine learning models for scalable Azure ML deployments.
- Conducted cost optimization by implementing auto-scaling and idle-resource cleanup, saving 20% monthly.

Walmart Global Tech | Senior Python Developer | Remote | Sep 2021 – Nov 2022

Application: Internal enterprise support chatbot and predictive analytics platform for IT operations.

Tech Stack: Azure OpenAI, Python, FastAPI, PGVector, Neo4j, XGBoost, PyTorch, Databricks, Docker, Kubernetes, Airflow

- Developed REST APIs and backend services in Python (FastAPI/Django) for incident management automation.
- Designed data ingestion and transformation pipelines using Airflow and Databricks for analytical reporting.
- Built AI-driven anomaly detection models using XGBoost and PyTorch to predict system outages.
- Integrated Azure OpenAI APIs for conversational support bots used by 10,000+ internal users.
- Designed PostgreSQL + PGVector schema for hybrid semantic search capabilities.
- Deployed AI models as containerized microservices using Docker and Kubernetes.
- Implemented CI/CD using Jenkins and GitLab CI to automate model retraining and service rollouts.
- Collaborated with data scientists to build scalable model evaluation pipelines ensuring production readiness.
- Conducted code reviews and performance tuning, reducing API latency by 25%.

Concentrix | Python Developer | San Jose, CA | Aug 2018 – Aug 2021

Application: CRM and analytics platform for call center operations and customer engagement.

Tech Stack: Python, Flask, FastAPI, PostgreSQL, AWS, Docker

- Developed and maintained microservices and RESTful APIs for CRM and analytics modules.
- Implemented ETL pipelines in Python for customer insights and KPI dashboards.
- Migrated on-premise systems to AWS, enabling faster data access and improved availability.
- Designed and optimized PostgreSQL queries, improving response time by 40%.
- Containerized applications using Docker for consistent deployment environments.
- Integrated AWS services (S3, Lambda, CloudWatch) for event-driven data workflows.
- Collaborated with BI teams to automate Power BI dashboards for executive reporting.
- Supported multi-environment CI/CD setup using Jenkins and Git.

Accenture | Software Developer | Milpitas, CA | Apr 2016 – Jun 2018

Application: Backend systems and analytics dashboards for finance and retail enterprise clients.

Tech Stack: Python, Django, Flask, Pandas, NumPy, PostgreSQL, Jenkins, AWS

- Developed and supported transaction monitoring applications using Django and Flask frameworks.
- Built data automation scripts in Python using Pandas and NumPy, cutting manual work by 50%.
- Integrated AWS services (EC2, S3) for data storage and deployment of backend applications.
- Supported CI/CD pipelines with Jenkins for seamless integration and delivery.
- Conducted unit and integration testing to maintain code quality and reduce production defects.
- Collaborated with cross-functional teams for agile sprint planning and release cycles.
- Enhanced application performance through query optimization and asynchronous processing.

EDUCATION: Bachelor's in Computer Science, 2012, University of Allahabad, India